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Engineering Applied Mathematics IIB for Two Players with Six Number of Jester Cards on Differential Calculus of Function of Several Variables

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Abstract

This work presents 16 number of cards from applied mathematics IIB degree level mainly focused on differential calculus of function of several variables. The jester cards are

A. Functions of several variables, graphs, and level curves.

B. Limits and continuity.

C. Partial and higher-order derivatives.

D. Directional derivatives and gradients.

E. Total differentials, tangent planes, and the chain rule.

F. Relative and absolute extrema, including Lagrange multipliers.

1 Introduction

At present, the decline in the reading community has reached a worrisome level. For many reasons, people have stopped sitting down and reading. For instance, wasting time on useless social media and in trivial places are among the many minor reasons. In 2025, Jenber proposed a solution to this through a major share company established under the title "Zero or One Trading Share Company." This share company believes that the problem mentioned above can be solved in the following ways. We found the solution included

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in this writing to be convincing (for more information, see the foundation document of 0 or 1 game [4]). The solution presented by this share company is briefly summarized below:

(1) Making reading or studying entertaining, profitable, and rewarding. One of the methods for studying or acquiring new knowledge is understanding things by associating them with their equivalents. For example, an English word has many synonyms. Trying to memorize them by associating them is considered one of the best methods. However, to make this exercise unforgettable, doing it in the form of a game while having fun and discussing in groups makes it preferable.There are scientific studies conducted around studying English vocabulary by preparing cards (for example, see [1, 2, 3, 5, 6, 7, 15, 16, 17, 18]).

(2) Understanding things by associating them according to their types stimulates thinking, makes one inquisitive, sparks discussions with friends, and even drives one to search for reference books. And when done this way, it takes a much longer time to forget that thing.

(3) Furthermore, if this association is done enjoyably in the form of a game with friends, and if a bonus/reward is given to the winner to encourage knowledge and learning, the game will become popular.

(4) Developing various types of entertaining games to execute the solutions mentioned in 1-3 above.

In this paper, we have prepared cards for the 0 or 1 game. We have briefly described the number of cards we prepared as follows. Let the number of jester cards be 6 and the number of players be $k = 2$. Therefore, the total number of cards, $n = 8 \times 2 + 6 = 22$. Thus, including the jester cards, there are a total of 22 cards. The player cards will be $22 - 6 = 16$ cards; 16 is a multiple of 8 (for more information, see the 0 or 1 game foundation document). The jester cards are as follows:

A. Functions of several variables, graphs, and level curves.

B. Limits and continuity.

C. Partial and higher-order derivatives.

D. Directional derivatives and gradients.

E. Total differentials, tangent planes, and the chain rule.

F. Relative and absolute extrema, including Lagrange multipliers.

2 Preliminary

Differential calculus of functions of several variables is an essential mathematical tool in engineering. Many engineering quantities depend on more than one independent variable. For example, temperature may depend on spatial coordinates, voltage may vary with

position, and the performance of an engineering system may depend on several design parameters.

This chapter introduces the principal concepts of multivariable differential calculus, including functions of several variables, limits and continuity, partial derivatives, gradients, directional derivatives, total differentials, tangent planes, the chain rule, and optimization. These concepts provide mathematical tools for modelling, analyzing, and optimizing engineering systems [8].

2.1 Definitions and Mathematical Examples

2.1.1 Functions of Several Variables

A function of several variables assigns a unique real number to each point in its domain. A function of two variables can be written as

$$f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}, \qquad f(x, y) = z.$$

For example, consider

$$f(x, y) = x^2 + 3xy + y^2.$$

At the point $(1, 2)$,

$$f(1, 2) = 1 + 3(1)(2) + 4 = 11.$$

The graph of a function of two variables is generally a surface in three-dimensional space. Level curves are obtained by setting $f(x, y) = c$, where c is a constant. These representations are useful for studying temperature distributions, voltage fields, and engineering performance functions [8].

2.1.2 Limits and Continuity

The limit of a function $f(x, y)$ as (x, y) approaches (a, b) describes the value approached by the function as the point (x, y) approaches (a, b).

For example,

$$\lim_{(x,y)\to(1,2)} (x^2 + y^2) = 5.$$

A function is continuous at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b),$$

where the limit and function value exist.

For a multivariable limit to exist, the function must approach the same value along every admissible path toward the point [11].

2.1.3 Partial Derivatives

A partial derivative measures the instantaneous rate of change of a function with respect to one variable, while the other variables are held constant.

For example, let

$$f(x, y) = x^2 y + 3xy^2.$$

Then

$$\frac{\partial f}{\partial x} = 2xy + 3y^2,$$

and

$$\frac{\partial f}{\partial y} = x^2 + 6xy.$$

Partial derivatives are used to investigate how changes in individual engineering variables affect the output of a system [13].

2.1.4 Higher-Order Partial Derivatives

Higher-order partial derivatives are obtained by differentiating partial derivatives again. For a sufficiently smooth function, the mixed partial derivatives satisfy

$$f_{xy} = f_{yx}.$$

For example, if

$$f(x, y) = x^2 y^3,$$

then

$$f_x = 2xy^3, \qquad f_y = 3x^2 y^2,$$

and consequently

$$f_{xy} = 6xy^2, \qquad f_{yx} = 6xy^2.$$

Second-order partial derivatives are also important in classifying critical points of functions [12, 13].

2.1.5 Directional Derivatives and Gradients

The gradient of a differentiable function of two variables is defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle .$$

The directional derivative in a unit direction $\mathbf{u} = \langle u_1, u_2 \rangle$ is

$$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \mathbf{u}.$$

For example, if

$$f(x, y) = x^2 + y^2,$$

then

$$\nabla f(1, 2) = \langle 2, 4 \rangle.$$

The gradient points in the direction of greatest instantaneous increase of the function, while the directional derivative measures the rate of change in a specified unit direction [9].

2.1.6 Total Differentials and Linear Approximations

For a differentiable function $z = f(x, y)$, the total differential is

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy.$$

For small changes in x and y, the change in the function can be approximated by

$$\Delta z \approx f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

For example, if

$$f(x, y) = x^2 + y^2,$$

then at $(1, 2)$,

$$dz = 2\,dx + 4\,dy.$$

This approximation is useful for estimating the effect of small measurement errors or design changes [8].

2.1.7 Tangent Planes

The tangent plane to the surface $z = f(x, y)$ at $(a, b, f(a, b))$ is

$$\begin{aligned} z - f(a, b) = {} & f_x(a, b)(x - a) \\ & + f_y(a, b)(y - b). \end{aligned}$$

For example, consider $z = x^2 + y^2$ at $(1, 2, 5)$. Since $f_x(1, 2) = 2$ and $f_y(1, 2) = 4$, the tangent plane is

$$z - 5 = 2(x - 1) + 4(y - 2),$$

or equivalently,

$$z = 2x + 4y - 5.$$

Tangent planes provide local linear approximations of smooth surfaces [8].

2.1.8 The Chain Rule

Suppose that $z = f(x, y)$, where $x = x(t)$ and $y = y(t)$. The chain rule gives

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

For example, let

$$z = x^2 + y^2, \qquad x = t, \qquad y = 2t.$$

Then

$$z = t^2 + 4t^2 = 5t^2,$$

and hence

$$\frac{dz}{dt} = 10t.$$

The chain rule is useful when engineering variables depend on time or on other intermediate variables [14].

2.1.9 Relative and Absolute Extrema

A relative maximum or minimum describes a local maximum or minimum of a function. An absolute maximum or minimum is the greatest or smallest function value over the entire specified domain.

For an interior critical point of a differentiable function, one commonly solves

$$f_x(x,y) = 0, \qquad f_y(x,y) = 0.$$

The second-derivative test uses the discriminant

$$D = f_{xx}f_{yy} - (f_{xy})^2.$$

At a critical point:

- If $D > 0$ and $f_{xx} > 0$, there is a local minimum.
- If $D > 0$ and $f_{xx} < 0$, there is a local maximum.
- If $D < 0$, the point is a saddle point.
- If $D = 0$, the test is inconclusive.

For example, $f(x,y) = x^2 + y^2$ has a global minimum at $(0,0)$, where $f(0,0) = 0$ [12].

2.1.10 Lagrange Multipliers

Lagrange multipliers provide a method for optimizing a function subject to a constraint. For an objective $f(x,y)$ and a constraint $g(x,y) = 0$, the equations are

$$\nabla f(x,y) = \lambda \nabla g(x,y), \qquad g(x,y) = 0.$$

For example, to optimize $f(x,y) = xy$ subject to

$$x^2 + y^2 = 1,$$

we solve

$$\langle y, x \rangle = \lambda \langle 2x, 2y \rangle, \qquad x^2 + y^2 = 1.$$

This method is useful in constrained engineering design and resource allocation [10].

3 Jester and Player Cards

3.1 Instructions

Each question has six fixed answer choices, labelled A, B, C, D, E, and F. Exactly three choices are correct for each question, and the other three are incorrect.

During each play, a student selects one answer choice. A correct selection earns a score of 1, while an incorrect selection earns a score of 0.

The mathematical task described in each question determines which three choices are correct. The same six answer-choice descriptions are used throughout the question set.

3.2 Jester Cards

A. Functions of several variables, graphs, and level curves.

B. Limits and continuity.

C. Partial and higher-order derivatives.

D. Directional derivatives and gradients.

E. Total differentials, tangent planes, and the chain rule.

F. Relative and absolute extrema, including Lagrange multipliers.

3.3 Player Cards

Question 1: Temperature Distribution

An engineer models the temperature distribution over a metal plate by

$$T(x, y) = 100 - x^2 - 2y^2.$$

Choose the correct concepts from options A–F that are relevant to representing this temperature field, calculating its partial derivatives, and determining its directional rates of change.

Question 2: Limits and Continuity

Consider

$$f(x, y) = \frac{x^2 - y^2}{x - y}, \qquad x \neq y.$$

Choose the correct concepts from options A–F that are relevant to representing the function, examining its limit near $(1, 1)$, and investigating the existence of a continuous extension.

Question 3: Partial Derivatives and Linearization

Given

$$f(x, y) = x^3y^2 + 4xy,$$

choose the correct concepts from options A–F that are relevant to representing the function, computing its partial derivatives, and constructing a local linear approximation.

Question 4: Voltage Field

An electrical engineer models voltage by

$$V(x, y) = x^2 + xy + y^2.$$

Choose the correct concepts from options A–F that are relevant to representing the voltage field, investigating its continuity, and determining its directional rates of change.

Question 5: Gradient and Directional Derivatives

Given

$$f(x, y) = x^2 + 3y^2,$$

choose the correct concepts from options A–F that are relevant to representing the function, computing its partial derivatives, and finding the direction of its greatest rate of increase.

Question 6: Total Differential

Suppose

$$z = f(x, y) = x^2 + y^2.$$

Choose the correct concepts from options A–F that are relevant to calculating partial derivatives, determining the gradient, and estimating small changes in the output using the total differential.

Question 7: Tangent Plane

A mechanical engineer studies the surface

$$z = x^2 + xy + y^2.$$

Choose the correct concepts from options A–F that are relevant to representing the surface, calculating its partial derivatives, and determining the tangent plane at a specified point.

Question 8: Time-Dependent Output

Let

$$z = x^2 + y^2, \qquad x = t, \qquad y = 2t.$$

Choose the correct concepts from options A–F that are relevant to computing partial derivatives, investigating directional rates of change, and using the chain rule to determine the time-dependent rate of change.

Question 9: Critical Points

Given

$$f(x, y) = x^2 + y^2,$$

choose the correct concepts from options A–F that are relevant to representing the function, calculating its partial derivatives, and locating and classifying its critical points.

Question 10: Constrained Optimization

A design parameter is chosen to minimize

$$f(x, y) = x^2 + 4y^2$$

subject to the constraint

$$x + y = 6.$$

Choose the correct concepts from options A–F that are relevant to representing the objective function, calculating the necessary derivatives, and solving the constrained optimization problem.

Question 11: Saddle Points

For

$$f(x, y) = x^2 - y^2,$$

choose the correct concepts from options A–F that are relevant to representing the function, calculating its second-order partial derivatives, and determining whether $(0, 0)$ is a saddle point.

Question 12: Lagrange Multipliers

A civil engineer models a performance measure by

$$f(x, y) = xy$$

subject to

$$x^2 + y^2 = 1.$$

Choose the correct concepts from options A–F that are relevant to representing the objective function, calculating the necessary derivatives, and finding the constrained extrema.

Question 13: Temperature Change

The temperature at a point in a solid is modeled by

$$T(x, y) = x^2 + 2y^2.$$

Choose the correct concepts from options A–F that are relevant to representing the temperature field, calculating its partial derivatives, and determining the rate of temperature change in a specified direction.

Question 14: Time-Dependent System

A system has output

$$z = xy,$$

where both x and y depend on time. Choose the correct concepts from options A–F that are relevant to calculating partial derivatives, determining directional rates of change, and applying the chain rule to the time-dependent output.

Question 15: Absolute Extrema

An engineer studies

$$f(x, y) = x^2 + xy + y^2$$

over a specified closed and bounded design region. Choose the correct concepts from options A–F that are relevant to representing the function, calculating its partial derivatives, and finding the largest and smallest values on the specified region.

Question 16: Constrained Engineering Design

An electrical engineer studies

$$P(x, y) = x^2 + 2xy + y^2$$

subject to

$$g(x, y) = x + y - 4 = 0.$$

Choose the correct concepts from options A–F that are relevant to representing the objective and constraint, calculating derivatives, and investigating constrained extrema.

Answer Key

The following table identifies the three intended correct concept choices for each question.

Question	Correct choices
1	A, C, D
2	A, B, C
3	A, C, E
4	A, B, D
5	A, C, D
6	C, D, E
7	A, C, E
8	C, D, E
9	A, C, F
10	A, C, F
11	A, C, F
12	A, C, F
13	A, C, D
14	C, D, E
15	A, C, F
16	A, C, F

Scoring rule:

$$\text{Score} = \begin{cases} 1, & \text{if the selected choice is correct,} \\ 0, & \text{if the selected choice is incorrect.} \end{cases}$$

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